Research on Financial Performance Evaluation of Agricultural Listed Companies Based on BP Neural Network Model

Liujuan Li
Yangpu Branch
Shanghai Open University
Shanghai, China
llj752018691@163.com

Abstract

This study aims to create a financial performance evaluation model for agricultural listed companies using the BP neural network, addressing traditional systems' inaccuracies. Based on the characteristics of these companies, a financial performance evaluation indicator system is constructed, with 42 A-share listed agricultural enterprises as the sample. The model includes an input layer with 17 neurons for key financial indicators, an optimized hidden layer, and an output layer for financial performance. Trained using the Levenberg-Marquardt algorithm, the model's performance is evaluated by MSE and R-squared metrics. Results show high accuracy, with MSE near zero and R^2 reaching one, indicating a strong fit between predictions and actual data. This BP neural network model offers a reliable tool for assessing agricultural enterprises' financial performance, providing valuable insights for management and decision-making.

CCS Concepts

• **General and reference** → Cross-computing tools and techniques; Evaluation.

Keywords

BP neural network, Data-Driven, performance evaluation

ACM Reference Format:
Liujuan Li. 2024. Research on Financial Performance Evaluation of Agricultural Listed Companies Based on BP Neural Network Model. In *2024 5th International Conference on Computer Science and Management Technology (ICCSMT 2024), October 18–20, 2024, Xiamen, China.* ACM, New York, NY, USA, 6 pages. https://doi.org/10.1145/3708036.3708187

1 Introduction

In the wave of globalization and technological innovation, Chinese agriculture confronts the dual challenges of transformation and upgrading. National policies vigorously promote agricultural modernization, emphasizing high-quality development and supply chain optimization, which places higher demands on agricultural listed companies to enhance their competitiveness and optimize their financial and operational performance. However, traditional financial evaluation systems struggle to comprehensively reflect the true value and market potential of agricultural enterprises.

This study innovatively introduces the BP Neural Network model, leveraging historical data to accurately predict financial performance and provide decision-makers with data-driven insights. The aim is to establish an evaluation system tailored to the characteristics of the agricultural industry, empowering agricultural listed companies, policymakers, and market participants to make precise policy decisions and jointly promote steady growth and a leap in competitiveness in the new era.

2 Literature Review

The research and application of BP (Back Propagation) neural networks have undergone extensive development. In the mid-1950s, the introduction of the M-P model laid the theoretical foundation for neural network research. In 1985, Rumelhart and McClelland[1] further elaborated on the concept of BP neural networks, highlighting their ability to construct multi-layer, non-linear complex networks through the error back-propagation algorithm. These networks possess robust mapping capabilities, allowing them to analyze and process large datasets to uncover patterns and logic, enabling scientific responses.

Odom and Sharda[2] pioneered the application of neural networks to bankruptcy prediction, initiating the use of complex algorithms for financial data processing. Subsequently, Tam and Kiang[3] demonstrated that neural networks outperformed traditional models in predicting bank failures, leading to the widespread application of BP neural networks in the financial sector.

Zhou Hong (2019)[4] utilized a GA-BP neural network model to establish a performance evaluation index system for listed companies, validating the model's feasibility through factor analysis and providing significant references for company management. Zhang Zhengang et al. (2020)[5] employed information entropy theory and BP neural networks to construct a performance evaluation model for listed white goods manufacturing enterprises, confirming the model's generalization capability and effectiveness. Cao Ye (2021)[6] analyzed the performance evaluation system of agricultural product circulation supply chains centered on supermarkets, using BP neural networks to establish the evaluation index system and conducting practical applications to identify ways to improve supply chain performance. Zeng Fanlong et al. (2022)[7] used the entropy weight–VIKOR algorithm and AGA-BP neural network model to construct a performance evaluation and prediction model for listed logistics companies in China, verifying its effectiveness and practicality with actual data. Zhang Junpeng et al. (2022)[8]

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
ICCSMT 2024, Xiamen, China
© 2024 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 979-8-4007-0999-9/24/10
https://doi.org/10.1145/3708036.3708187

Table 1: Financial Performance Evaluation Indicator System for Listed Agricultural Companies.

Primary Indicators	Secondary Indicators	Code	Indicator Definition
Solvency	Current Ratio	X1	Current Assets / Current Liabilities
	Quick Ratio	X2	(Current Assets - Inventory) / Current Liabilities
	Debt to Asset Ratio	X3	Total Liabilities / Total Assets
Profitability	Return on Assets (ROA)	X4	Net Profit / Average Total Assets
	Return on Equity (ROE)	X5	Net Profit / Average Shareholders' Equity
	Operating Profit Margin	X6	Operating Profit / Operating Revenue
Growth Ability	Total Asset Growth Rate	X7	(Total Assets at End of Current Period - Total Assets at Beginning of Current Period) / Total Assets at Beginning of Current Period
	Net Profit Growth Rate	X8	(Net Profit of Current Period - Net Profit of Same Period Last Year) / Net Profit of Same Period Last Year
	Sustainable Growth Rate	X9	Return on Equity × Retained Earnings Ratio / (1 - Return on Equity × Retained Earnings Ratio)
Operating Ability	Accounts Receivable Turnover	X10	Operating Revenue / Ending Balance of Accounts Receivable
	Inventory Turnover	X11	Cost of Goods Sold / Average Inventory
	Current Asset Turnover	X12	Operating Revenue / Average Current Assets
	Non-Current Asset Turnover	X13	Operating Revenue / Average Net Non-Current Assets
	Total Asset Turnover	X14	Operating Revenue / Average Total Assets
Cash Ability	Net Profit Cash Content	X15	Net Cash Flow from Operating Activities / Net Profit
	Operating Index	X16	Net Cash Flow from Operating Activities / Gross Cash Flow from Operating Activities
Shareholder Profitability	Earnings Per Share	X17	Net Profit / Total Shares at End of Period

selected financial data from 18 listed cross-border e-commerce companies, constructing a performance evaluation system based on BP neural networks, which showed good fitting and generalization capabilities. Bu Wei et al. (2023)[9] constructed a university performance evaluation index system based on the "input-output" theory and analyzed the scientific research performance and influencing factors of universities in Jiangsu Province using AHP and GA-BP neural networks. The results indicated an overall increase in university research performance, with a hierarchical pattern among different types of universities and significant polarization.

While BP neural networks have demonstrated broad potential in the realm of financial performance research, there remains a significant gap in their application within the financial assessment of agricultural listed companies. This study introduces a BP neural network model designed to address this void. By leveraging historical financial data to forecast corporate performance and employing an optimized neural structure to enhance predictive accuracy, the model offers a novel perspective for evaluating the financial health and market potential of agricultural enterprises, thereby equipping decision-makers with data-informed insights.

3 Establishment of Financial Performance Indicator System

3.1 Sample Selection and Data Sources

The financial data utilized in this study is sourced from the China Stock Market & Accounting Research (CSMAR) database provided by Guotaian Financial Data Service Center. The sample consists of 42 agricultural enterprises listed on the A-shares market from 2021 to 2023, encompassing industries such as planting, breeding, and fisheries.

3.2 Selection of Financial Performance Evaluation Indicators

A robust financial performance indicator system is vital for improving evaluation models' effectiveness and accuracy. Our study's system, detailed in Table 1, balances theoretical foundation with practicality, prioritizing indicator frequency in literature and relevance to issues addressed.

4 Introduction to BP Neural Networks

4.1 Network Structure

The BP (Back Propagation) neural network, a key multilayer feed-forward model in artificial intelligence, simulates neural activity to manage nonlinear relationships and is used in various applications like pattern recognition and system prediction. Comprising an

input layer, hidden layers, and an output layer with interconnected neurons, the BP network learns and memorizes input-output mappings without prior mathematical formulation. Recognizing that financial performance is influenced by a multitude of indicators across different dimensions, this study utilizes the BP neural network to model the intricate relationships between these indicators and the financial performance.

4.2 Algorithm Flow

4.2.1 Forward Propagation. In the forward propagation process, the input signal is transmitted from the input layer. In this context, the input vector $x = [x_1, x_2, ..., x_{17}]^T$ represents the 17 indicators extracted earlier. Through the hidden layer, these signals ultimately reach the output layer, yielding the network's predicted output, which is the company's financial performance value. The input to the neuron in the l-th layer is:

$$z_l = W_l^T a_{l-1} + b_l \tag{1}$$

Where W_l is the weight matrix for the l-th layer, b_l is the bias vector, and a_{l-1} is the activated output vector of the (l-1)-th layer. The output after the activation function is applied is:

$$a_l = f\left(z_l\right) \tag{2}$$

Where $f(z_l)$ represents the activation function. The final output is denoted as $y = a_L$.

4.2.2 Backward Propagation. In the backward propagation phase, the prediction error of the network is propagated back from the output layer to update the weights and biases. The loss function of the BP neural network constructed in this paper is the Mean Squared Error (MSE) loss function, denoted as *MSELoss(y,t)*, where y represents the network output and t represents the desired output. The error of the output layer is calculated using the following formula:

$$\delta_L = \nabla_\alpha E|_L \cdot f^{'}\left(z_L\right) \tag{3}$$

In this context, $\nabla_\alpha E|_L$ represents the gradient of the loss function with respect to the activation of the output layer, and $f^{'}(z_L)$ is the derivative of the activation function. The error of the hidden layers is calculated as follows:

$$\delta_l = \left(W_{l+1}\delta_{l+1}\right) \cdot f^{'}\left(z_l\right) \tag{4}$$

Where δ_{l+1} is the error term from the next layer. Then, the weights and biases are updated using the formulas: $\Delta W_l = -\eta a_{l-1}\delta_l^T$ *and* $\Delta b_l = -\eta\delta_l$, where η is the learning rate. This entire process is iterated multiple times until the prediction error of the network reaches a predefined threshold or other stopping criteria are met.

5 Empirical Analysis

5.1 Data Preprocessing

Prior to BP neural network input, preprocessing entails data cleaning and imputation of missing values. Our empirical analysis utilizes financial data from 42 agricultural listed firms (2021-2023), selecting 17 key indicators. SPSS identified missing values in net profit growth, receivables turnover, EPS, net cash content, and operating index. Leveraging regression for imputation, considering indicator correlations.

Additionally, normalization (max-min to [0,1]) is applied to all samples, mitigating dimensional differences for BP network compatibility.

5.2 Construction of BP Neural Network

This paper primarily utilizes MATLAB R2023a as the simulation platform to construct the BP neural network.

The input layer of the BP neural network consists of 17 neurons, each corresponding to one of the 17 selected indicators.

For the hidden layer, a range of neuron counts can be suggested using Equation 5 as a reference, and then the optimal number of hidden layer neurons can be identified through comparative training.

$$m = \sqrt{n + l} + \alpha \tag{5}$$

Equation 5 defines *m, n, l* as neurons in hidden, input, output layers. α ranges from 0 to 10. Our calculations set the optimal hidden layer neurons for this BP neural network at 14.

The output layer represents the company's financial performance, and the desired output is obtained through the entropy weight method.

The steps to calculate the desired output using the entropy weight method are as follows:

Step 1: Data Preprocessing. As standardization of indicators has been completed in the previous section, the current task is to transform negative indicators into positive ones to ensure data consistency.

Step 2: Data Translation. To eliminate the influence of zeros and negative values without compromising data usability, this paper translates the entire dataset using Equation 6, with a determined value of 0.0001 units.

$$Y_{ij}^{'} = X_{ij}^{'} + \beta \tag{6}$$

Step 3: Calculate the Information Entropy of Indicators, as shown in Equation 7.

$$\begin{aligned} p_{ij} &= \frac{Y_{ij}^{'}}{\sum_{k=1}^{m} Y_{kj}^{'}} \\ e_j &= -\sum_{i=1}^{m} p_{ij}\log(p_{ij}) \end{aligned} \tag{7}$$

Step 4: Determine the Indicator Weights and Comprehensive Score, as shown in Equation 8. The comprehensive score is the weighted sum of all indicators.

$$\begin{aligned} d_j &= (1 - e_j) \Big/ \sum_{k=1}^{n} (1 - e_k) \\ S_i &= \sum_{j=1}^{n} d_j Y_{ij}^{'} \end{aligned} \tag{8}$$

Through the above steps, we can obtain the financial performance score for each sample, which will serve as the desired output for the BP neural network. Therefore, the number of output layer neurons in the BP neural network constructed in this paper is 1.

5.3 BP Neural Network Training

To train and test the BP neural network, data's divided annually: 2021-2022 for training (84 samples), 2023 for testing (42 samples). Using Matlab R2023a, we train the network with the Levenberg-Marquardt algorithm.

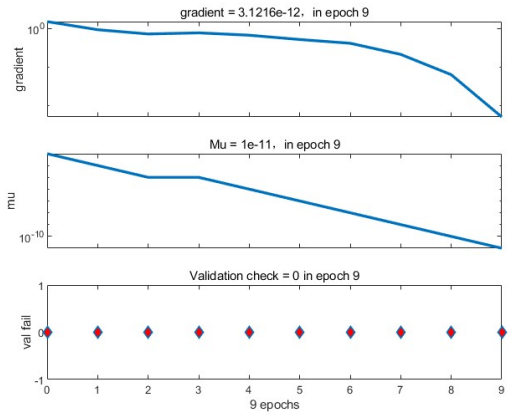

Figure 1: Illustration of the Training Process.

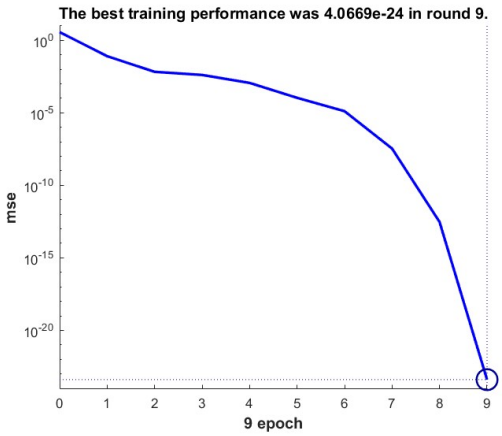

Figure 2: Performance of the Model on MSE during Training Process.

As can be seen from Figure 1, at the ninth iteration, the gradient and Mu value both reach their minimums, indicating that the model has learned the optimal weights at this point and training can be stopped. Additionally, the performance of the model during the training process in terms of MSE and R^2 is visualized in Figure 2 and Figure 3, respectively.

From Figure 2, it can be observed that by the 9th iteration, the MSE of the model on the training set approaches 0. Additionally, Figure 3 shows that at the 9th iteration, the training data and the fitted function lie on the same straight line, indicating that the R-squared (R^2) value equals 1 at this point.

In Figure 4, the yellow line represents an error of 0. It can be observed that more than half of the samples' errors are close to the yellow line. Meanwhile, the horizontal axis of the figure indicates that the error range covers both negative and positive values, but most of them are concentrated near zero, suggesting that the model's prediction deviations are relatively small.

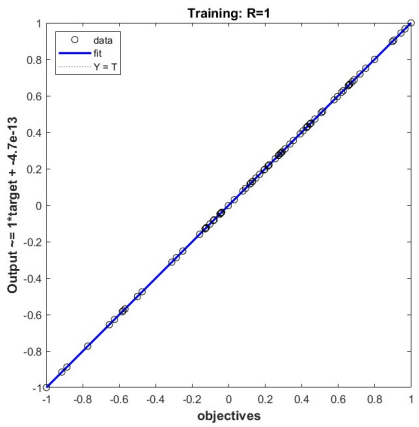

Figure 3: Performance of the Model on R^2 during Training Process.

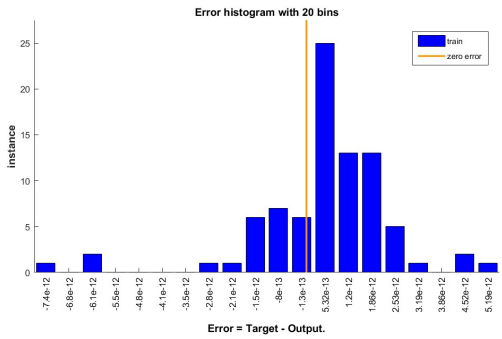

Figure 4: Visualization of Error Histogram.

In conclusion, the BP neural network constructed in this paper exhibits excellent performance on the training set, demonstrating a significant effect in fitting the company's financial performance.

5.4 BP Neural Network Testing

We will further validate the effectiveness of the BP neural network in fitting the company's financial performance on the test set. Figure 5 demonstrates the relationship between the actual values and predicted values by the model on both the training set and the validation set.

On the test set, it is evident that there exists a linear relationship between the actual values and the predicted values. Furthermore, to gain a more intuitive understanding of the errors between the actual values and predicted values for each test sample, this paper visualizes this relationship, as depicted in Figure 6. It can be observed that the predicted values are consistent with the actual values for the majority of the samples.

Table 2 indicates that the neural network training results are close to the actual values, with controllable errors, demonstrating excellent training performance and successful model validation.

Table 2: 2023 Agri-Firms' Expected vs. Predicted Financial Perf.

Company	expected value	predicted value	Error value	Company	expected value	predicted value	Error value
Pingtan Development	0.319248	0.336115	0.000284	Lihua Stocks	0.398332	0.352339	0.002115
Fengle Seeds	0.423172	0.423031	0	Xiaoming Stocks	0.309739	0.280933	0.00083
Luoniushan	0.286847	0.288431	0.000003	Hualv Bio-Tech	0.442648	0.441855	0.000001
China National Fisheries Corporation	0.391112	0.390667	0	Wanchen Group	0.438524	0.376407	0.003859
Longping High-Tech	0.372966	0.384064	0.000123	Kaichuang International	0.450581	0.463148	0.000158
Dongrui Stocks	0.245758	0.273705	0.000781	Yasheng Group	0.382783	0.393171	0.000108
Denghai Seeds	0.497568	0.517088	0.000381	Dahu Stocks	0.38258	0.38329	0.000001
Minhe Stocks	0.283883	0.284745	0.000001	Nongfa Seeds	0.464613	0.473369	0.000077
St. Claire Foods	0.410324	0.404434	0.000035	Dunhuang Seeds	0.402994	0.426479	0.000552
Yisheng Stocks	0.479019	0.467194	0.00014	Xinjiang New Agri Development	0.396553	0.413606	0.000291
Fujian Jinsen	0.40603	0.407788	0.000003	Wanxiang Denagriculture	0.447808	0.466974	0.000367
Baiyang Stocks	0.388314	0.389802	0.000002	Haodangjia	0.361155	0.358596	0.000007
Muyuan Foods	0.34127	0.312123	0.00085	Xinsai Stocks	0.301955	0.329269	0.000746
Zhongxing Mushroom Industry	0.39239	0.400064	0.000059	Beidahuang	0.600222	0.535721	0.00416
Xiangjia Stocks	0.348118	0.329712	0.000339	Xinwufeng	0.222421	0.219622	0.000008
Zhonglu B	0.416642	0.419482	0.000008	Hainan Rubber Group	0.419978	0.396855	0.000535
Quanyin High-Tech Seeds	0.411188	0.443479	0.001043	Juxing Agriculture & Husbandry	0.266154	0.265905	0
Western Animal Husbandry	0.384566	0.38219	0.000006	Shennong Group	0.350718	0.340682	0.000101
Shennong Science and Technology	0.448087	0.420972	0.000735	Qiule Seeds	0.491417	0.514245	0.000521
Wens Foods	0.307291	0.287478	0.000393	Knights Dairy	0.417063	0.430388	0.000178
Snowmushroom Biotech	0.31448	0.299646	0.00022	Kangnong Seeds	0.464207	0.493209	0.000841

6 Conclusion

This paper meticulously selects 17 financial indicators across six core dimensions: debt repayment, profitability, growth, operational efficiency, cash flow, and shareholder profitability, to establish a comprehensive financial performance evaluation system tailored for leading agricultural industrialization enterprises. Subsequently, employing the entropy weight method in conjunction with the BP neural network model, the system's applicability and accuracy are rigorously calculated and validated. Following exhaustive training and testing, the BP neural network-based evaluation model demonstrates remarkable accuracy and practicality in real-world applications, providing robust data support and insightful decision-making references for agricultural enterprises' financial assessment and management. As such, it emerges as an ideal tool for evaluating the financial performance of agricultural enterprises. Further research will explore a variety of neural network architectures, such as deep learning models, to enhance the accuracy of predictions.

References

[1] Rumelhart, D. E., McClelland, J. L., & the PDP Research Group. 1987. Parallel Distributed Processing: Explorations in the Microstructure of Cognition. MIT Press, Cambridge, MA.

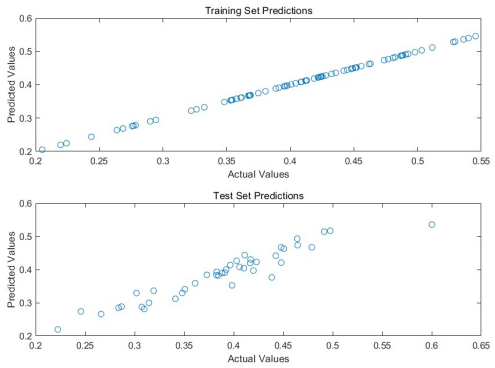

Figure 5: visualizes the relationship between the predicted values and the actual values.

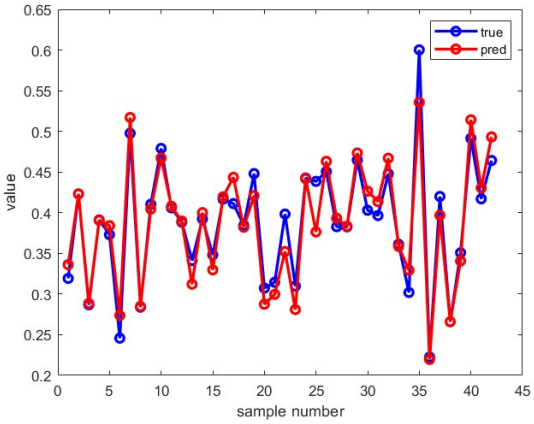

Figure 6: Comparison of Predicted Values and Actual Values on the Test Set.

[2] Odom, M. D., & Sharda, R., "A Neural Network Model for Bankruptcy Prediction," in 1990 IJCNN International Joint Conference on Neural Networks, IEEE, pp. 163–168, July 1990.
[3] Tam, K. Y., & Kiang, M. Y., "Managerial Applications of Neural Networks: The Case of Bank Failure Predictions," Management Science, vol. 38, no. 7, pp. 926–947, July 1992.
[4] Zhou Hong. 2019. Application of GA-BP Neural Network Model in Corporate Performance Evaluation. Finance and Accounting Communication, (17), 31-36.
[5] Zhang Zhengang, Xie Mengxin, and Lin Dan. 2020. Performance Evaluation System for Manufacturing Enterprises in China Based on BP Neural Network: A Case Study of Listed White Goods Manufacturing Enterprises. Science and Technology Management Research, 40(15), 217-223.
[6] Cao Ye. 2021. Research on Performance Evaluation of Agricultural Product Circulation Supply Chain Based on BP Neural Network. Commercial Economic Research, (11), 134-137.
[7] Zeng Fanlong, Ni Jing, and Wang Yuhua. 2022. Research on Performance Evaluation of Listed Logistics Enterprises Based on Entropy Weight–VIKOR and AGA-BP Models: A Case Study of China's Listed Logistics Enterprises. Journal of University of Shanghai for Science and Technology, 44(01), 94-102.
[8] Zhang Junpeng, Xu Zhixiang, and Tao Chunfeng. 2022. Performance Evaluation System of Listed Cross-border E-commerce Enterprises Based on BP Neural Network. China Foreign Investment, (18), 74-76.
[9] Bu Wei, Yu Nannan, and Chen Junbing. 2023. Performance Evaluation and Influencing Factors Analysis of Universities in Jiangsu Province Based on AHP and GA-BP Neural Network Analysis. Jiangsu Science & Technology Information, 40(03), 64-70.